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YOU SCRATCH MY BACK, I'LL RIDE ON YOURS

We have considered parental, sexual, and aggressive interactions between survival machines belonging to the same species. There are striking aspects of animal interactions which do not seem to be obviously covered by any of these headings. One of these is the propensity that so many animals have for living in groups. Birds flock, insects swarm, fish and whales school, plains-dwelling mammals herd together or hunt in packs. These aggregations usually consist of members of a single species only, but there are exceptions. Zebras often herd together with gnus, and mixed-species flocks of birds are sometimes seen.

The suggested benefits that a selfish individual can wrest from living in a group constitute rather a miscellaneous list. I am not going to trot out the catalogue, but will mention just a few suggestions. In the course of this I shall return to the remaining examples of apparently altruistic behaviour that I gave in Chapter 1, and which I promised to explain. This will lead into a consideration of the social insects, without which no account of animal altruism would be complete. Finally in this rather miscellaneous chapter, I shall mention the important idea of reciprocal altruism, the principle of 'You scratch my back, I'll scratch yours'.

If animals live together in groups their genes must get more benefit out of the association than they put in. A pack of hyenas can catch prey so much larger than a lone hyena can bring down that it pays each selfish individual to hunt in a pack, even though this involves sharing food. It is probably for similar reasons that some spiders cooperate in building a huge communal web. Emperor penguins conserve heat by huddling